

BIKRANT NATH

✉ bikrantnath4@gmail.com ☎ +91 9101121475 📄 [Portfolio](#) [in /bikrant-nath](#) [bikrantnath22](#)

Guwahati, Assam

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and Full-Stack Developer with expertise in privacy-preserving machine learning, Generative AI, LLM-powered applications, and scalable system design. Holds M.Tech in Computer Science & Engineering from Tezpur University (2026) with hands-on research experience in differentially private synthetic data generation. Proficient in delivering efficient, maintainable, and high-quality code across AI/ML and web/mobile stacks, with strong problem-solving skills and a passion for emerging technologies.

EDUCATION

M.Tech — Computer Science & Engineering | Tezpur University 2024 – 2026 · Tezpur, Assam
CGPA: 7.28 / 10 (till 3rd semester)

MCA — Master of Computer Applications | Tripura University 2022 – 2024 · Agartala, Tripura
CGPA: 7.71 / 10

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, C#, HTML, CSS

Frameworks & Libraries: React, Next.js, React Native, Node.js, Express.js, Tailwind CSS, Firebase, PyTorch, Opacus

AI / ML: Machine Learning, Deep Learning, Neural Networks, Supervised & Unsupervised Learning, NLP, GANs, Differential Privacy, Normalizing Flows, LLMs, RAG, Prompt Engineering, AI Agents, Embeddings, Hyperparameter Tuning, Data Preprocessing

Databases: MongoDB, SQL

Tools & Platforms: Git, GitHub, VS Code, Postman, Unity, WebSockets

Cybersecurity: Penetration Testing, Ethical Hacking Methodology, Nmap, Wireshark, Burp Suite, Metasploit, SQLmap

AWARDS & ACHIEVEMENTS

🏆 **Codestellation Hackathon First Runner-Up** — Assam Engineering College (AEC) · 2022

Built a resume builder web application in a competitive hackathon setting — featured template selection, real-time preview, and one-click PDF export.

EXPERIENCE

Software Development Intern | Fineworks Media & Solution Technologies Jan 2022 – Jul 2022 · Assam

- Built a full-stack hotel booking platform from scratch (React frontend + Node.js/Express backend + MongoDB), reducing manual booking errors through automated availability checks and reservation workflows.
- Designed and implemented RESTful APIs for booking management, inventory tracking, and reservation workflows — supporting concurrent room lookups and live status updates.
- Integrated a third-party online payment gateway, enabling secure end-to-end transactions with confirmation receipts.

PROJECTS

DP-CTGAN with Rare Minority Sampling 2025–2026

Privacy-Preserving Synthetic Data Generation |

- Designed and implemented a Differentially Private CTGAN pipeline using PyTorch and Opacus, evaluated across multiple privacy budget configurations on four real-world datasets with formal (ϵ, δ) -DP guarantees tracked via the Rényi Differential Privacy accountant
- Integrated StableNVP normalizing flow scoring and gradient refinement to improve sample fidelity — referenced novel flow-based approaches (Lee et al., arXiv:2412.19543) for diversity-aware synthesis.

- Benchmarked against two established DP baselines — PATE-GAN (SynCity) and DP-CGAN (dp-cgans) — across all four datasets; DP-CTGAN combined consistently outperformed both on AUC and distributional fidelity (KS),
- Implemented MIA (Membership Inference Attack) privacy auditing with KNN-MIA and LR-MIA; confirmed attack scores near 0.50 across all configurations, empirically validating privacy preservation.
- Tuned hyperparameters with Optuna using a three-stage search (No-DP GAN, DP GAN per ϵ , and rare-sampling/flow components) with a 3-way train/val/test split to prevent data leakage; evaluated using TSTR (RF and KNN classifiers) reporting AUC and AUPRC, distribution metrics (KS, JS divergence), and dual MIA auditing (KNN-MIA and LR-MIA) with a sealed holdout to prevent evaluation leakage.

Tech: PyTorch, Opacus, Optuna, Scikit-learn, Pandas, NumPy, CTGAN, Normalizing Flows, Python

Learning-based Cognitive Assessment using Virtual Reality

2024

- Designed a VR-based computer literacy learning and assessment system for school students, accessible via Meta Quest, with a companion mobile/desktop app for teacher-side progress tracking.
- Developed assessment modules to deliver quizzes, collect responses, and track per-student performance — 3D VR interface simulated computer hardware interactions (keyboard, motherboard, peripherals).
- Conducted controlled experiment comparing VR vs. text-based learning groups; VR group demonstrated measurably higher engagement and quiz comprehension scores.
- Proposed a hybrid pedagogy model combining text-based theory with VR practical simulations — documented usability testing results and customization roadmap.

Tech: Unity, C#, Meta Quest SDK, React Native (companion app), Node.js, MongoDB

AI Chat Agent — Human-like Messaging System

2025

- Built an AI-powered conversational agent that replicates individual messaging styles using LLM-based generation with persistent memory/context handling.
- Designed the system to learn from prior conversations and adapt tone, vocabulary, and response cadence to match a target user's communication patterns.
- Implemented real-time message delivery using Node.js + WebSockets

Tech: Node.js, WebSockets, LLM API, Prompt Engineering, Memory/Context Management, MongoDB

Animal Rescue Platform

2024

- Built a full-stack NGO-facing rescue coordination platform with React.js, Node.js (WebSockets), MongoDB, and Cloudinary image uploads.
- Implemented real-time geolocation-based NGO matching and alert dispatch, reducing manual coordination overhead for rescue case intake.
- Designed an admin dashboard with case tracking, status analytics, and NGO management — supporting multi-role access.

Tech: React.js, Node.js, WebSockets, MongoDB, Cloudinary, Geolocation API

Athaana — eCommerce (Startup Nalbari, Assam based)

2022

- Developed a production full-stack grocery eCommerce platform for a local startup — product listings, cart, order management, and live order tracking.
- Implemented user authentication, checkout flow, and order tracking system serving real customers in Nalbari.

Tech: React, Node.js, Express.js, MongoDB, Tailwind CSS

Vastramaniala — Shopify Clothing Store

2024

- Customized a Shopify storefront for an online clothing brand — implemented Liquid templates for product pages, collections, and UI enhancements.

Tech: Shopify, Liquid Templates, JavaScript, CSS